



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to GK FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

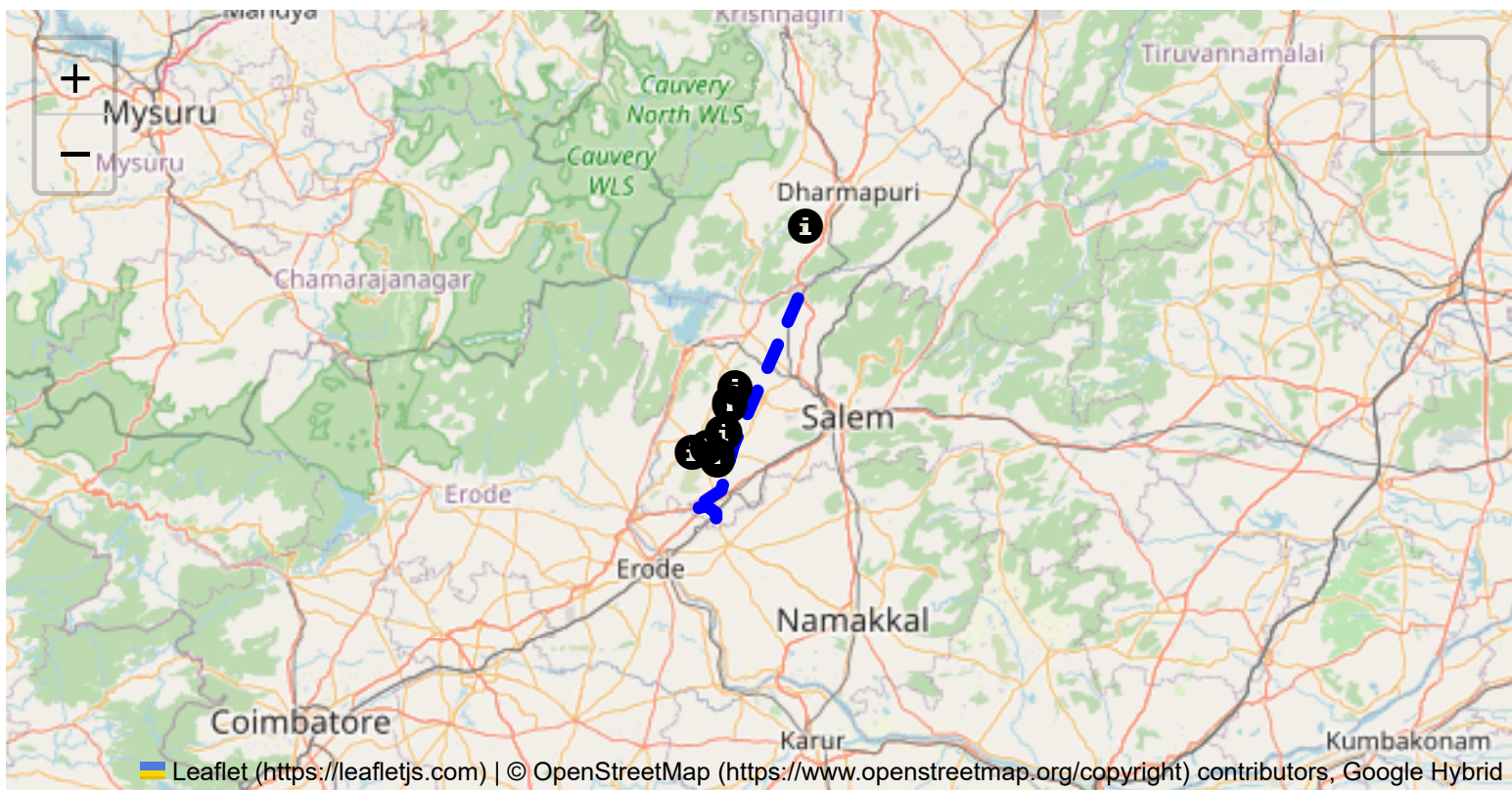
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

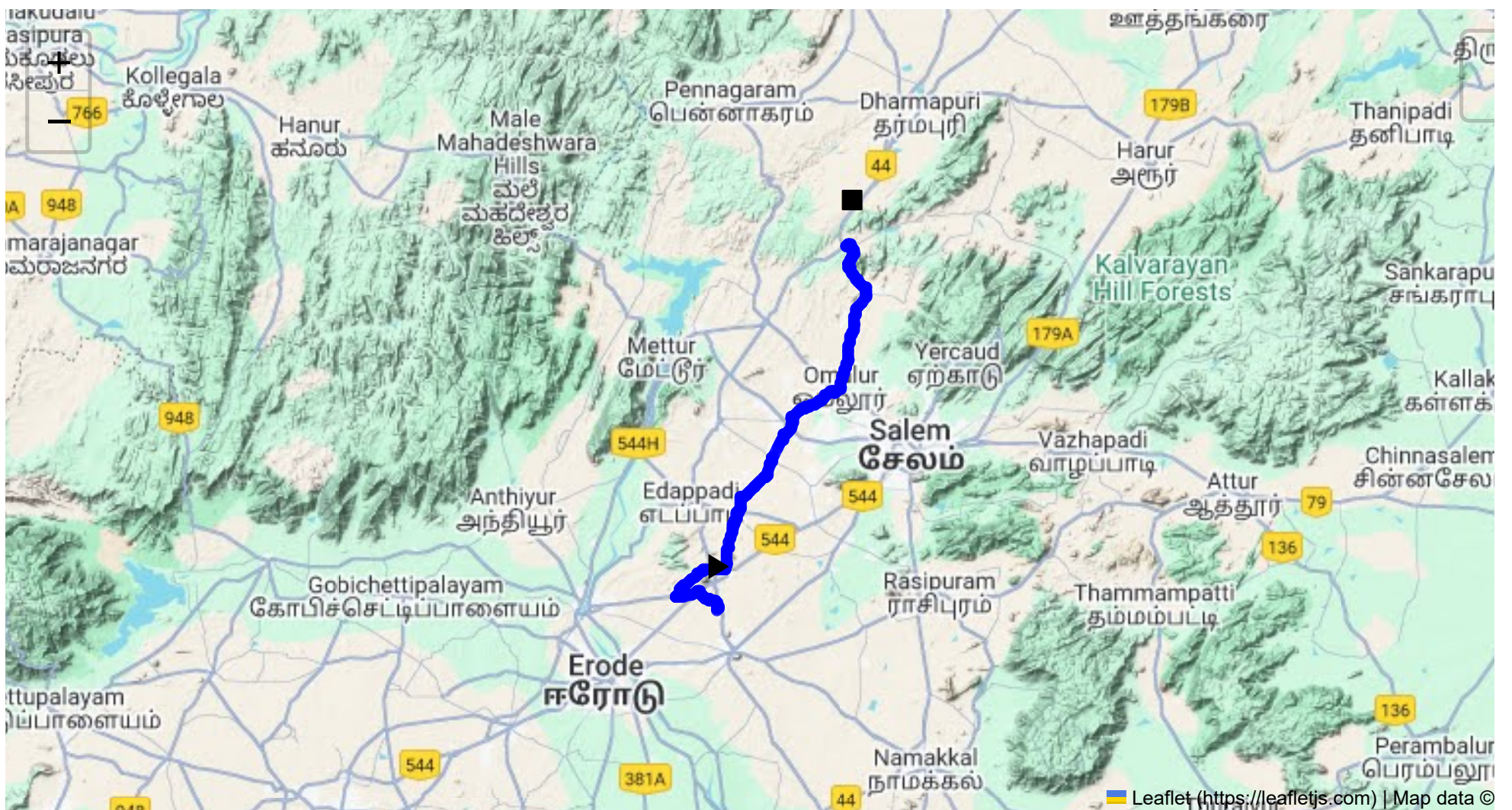
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 79.19 km
Estimated Duration: 1.8 hours
Adjusted Duration (Heavy Vehicle): 2.2 hours
Start: (11.4381, 77.8734)
End: (11.93939, 78.061266)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W to W3Q6+M55 spans approximately 79.19 kilometers across Tamil Nadu, navigating through several key points such as Sankari RS, Padaiveedu, Ayveli, Erumaipatti, and Koranampatti. It primarily uses Sankari RS - Erode Privu Road and SH 86, two significant arterial roads in the region, facilitating travel through a mix of urban and semi-rural environments.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu typically experiences tropical weather with distinct wet (June to October) and dry (November to May) seasons. The monsoon rains can cause waterlogging and decreased visibility, increasing the risk of accidents. Flash floods, although rare, can create hazardous conditions, especially on SH 86.

3. Analysis of Traffic Patterns

- **Peak Hours:** Morning (8-10 AM) and evening (4-6 PM).
- **Congestion-Prone Areas:** Sankari RS area and especially near major junctions along the Bhavani Main Road and SH 86.
- **Traffic Flow:** Generally smooth but expect slowdowns during peak hours, primarily due to local commuter traffic and market activity.

4. Assessment of Road Quality and Infrastructure

The road quality varies along the route:

- **Sankari RS - Erode Privu Road:** Generally well-maintained with occasional potholes.
- **SH 86:** Mixed conditions; better near urban areas but deteriorates towards the rural stretches.
- Infrastructure such as signage and lighting can be inconsistent, particularly in rural sections.

5. Suggestions for Alternative Routes for Emergencies

In emergencies, detouring to the Chennai-Bangalore Highway (NH44) if accessible, or using local roads connecting to Erode or Salem for medical emergencies, can be considered although these routes may add distance.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Transport of hazardous materials is subject to regulations like obtaining proper permits, using correct signage, and observing allowed transit times mainly during off-peak hours to minimize risk.

7. Overview of Historical Incidents

Historical data indicates occasional vehicle breakdowns and accidents involving heavy vehicles primarily due to human error or mechanical failures. No significant incidents involving hazardous materials have been reported on this route.

8. Environmental Considerations and Sensitive Areas

- **Ecological Zones:** Certain stretches near Erumaipatti and Koranampatti may be close to agrarian landscapes.

- **Respect for local fauna and flora is essential**, particularly when transporting hazardous materials that could potentially impact these areas.

9. Analysis of Communication Coverage

Mobile network coverage is generally reliable in urban and semi-urban areas but can be patchy in more rural segments, especially between Ayveli and Erumaipatti.

10. Estimated Emergency Response Times

- **Urban Segments (Sankari, Koranampatti):** Approximately 20-30 minutes.
- **Rural Segments (Between Ayveli and Erumaipatti):** 30-50 minutes depending on time of day and weather conditions.

12. Overall Summary of Risk Assessment

The route poses moderate risks which can be mitigated by:

- Adhering to traffic rules and transporting materials within designated off-peak times.
- Remaining vigilant during monsoon conditions and on less maintained road stretches.
- Ensuring proper communication equipment is on hand to manage potential connectivity issues.
- Monitoring traffic updates to anticipate areas of slow movement and adjust junction approaches accordingly.

Overall, with careful planning and adherence to safety protocols, this route can be managed efficiently for the transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	Medium	11.44900, 77.87406	30 KM/Hr
3	Turn	Medium	11.45357, 77.85789	30 KM/Hr
4	Turn	High	11.45352, 77.85698	15 KM/Hr
5	Turn	Medium	11.45847, 77.85140	30 KM/Hr
6	Turn	Medium	11.45898, 77.85135	30 KM/Hr
7	Turn	Medium	11.46303, 77.84945	30 KM/Hr
8	Turn	Medium	11.46314, 77.84928	30 KM/Hr
9	Turn	High	11.45542, 77.81725	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	High	11.45634, 77.81673	15 KM/Hr
11	Turn	Medium	11.49416, 77.88011	30 KM/Hr
12	Turn	Medium	11.49421, 77.88021	30 KM/Hr
13	Turn	High	11.49383, 77.88526	15 KM/Hr
14	Turn	High	11.49395, 77.88537	15 KM/Hr
15	Turn	Medium	11.49465, 77.88488	30 KM/Hr
16	Turn	Medium	11.56096, 77.89931	30 KM/Hr
17	Turn	High	11.57035, 77.90223	15 KM/Hr
18	Turn	High	11.57041, 77.90221	15 KM/Hr
19	Turn	Medium	11.58279, 77.90658	30 KM/Hr
20	Turn	Medium	11.63833, 77.94818	30 KM/Hr
21	Turn	Medium	11.64171, 77.95184	30 KM/Hr
22	Turn	Medium	11.64243, 77.95198	30 KM/Hr
23	Turn	Medium	11.64721, 77.95118	30 KM/Hr
24	Turn	Medium	11.67858, 77.96834	30 KM/Hr
25	Turn	Medium	11.68915, 77.97734	30 KM/Hr
26	Turn	High	11.70227, 77.97864	15 KM/Hr
27	Turn	Medium	11.71921, 78.01133	30 KM/Hr
28	Turn	High	11.73651, 78.03878	15 KM/Hr
29	Turn	High	11.93670, 78.05606	15 KM/Hr
30	Turn	Medium	11.93877, 78.05736	30 KM/Hr
31	Turn	Medium	11.93965, 78.05742	30 KM/Hr
32	Turn	Medium	11.94007, 78.05886	30 KM/Hr
33	Turn	High	11.93896, 78.06064	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
14	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
15	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
16	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
18	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
19	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
28	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45634, 77.81673



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49416, 77.88011



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49421, 77.88021



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49383, 77.88526



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49395, 77.88537



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49465, 77.88488



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56096, 77.89931



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57035, 77.90223



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.57041, 77.90221

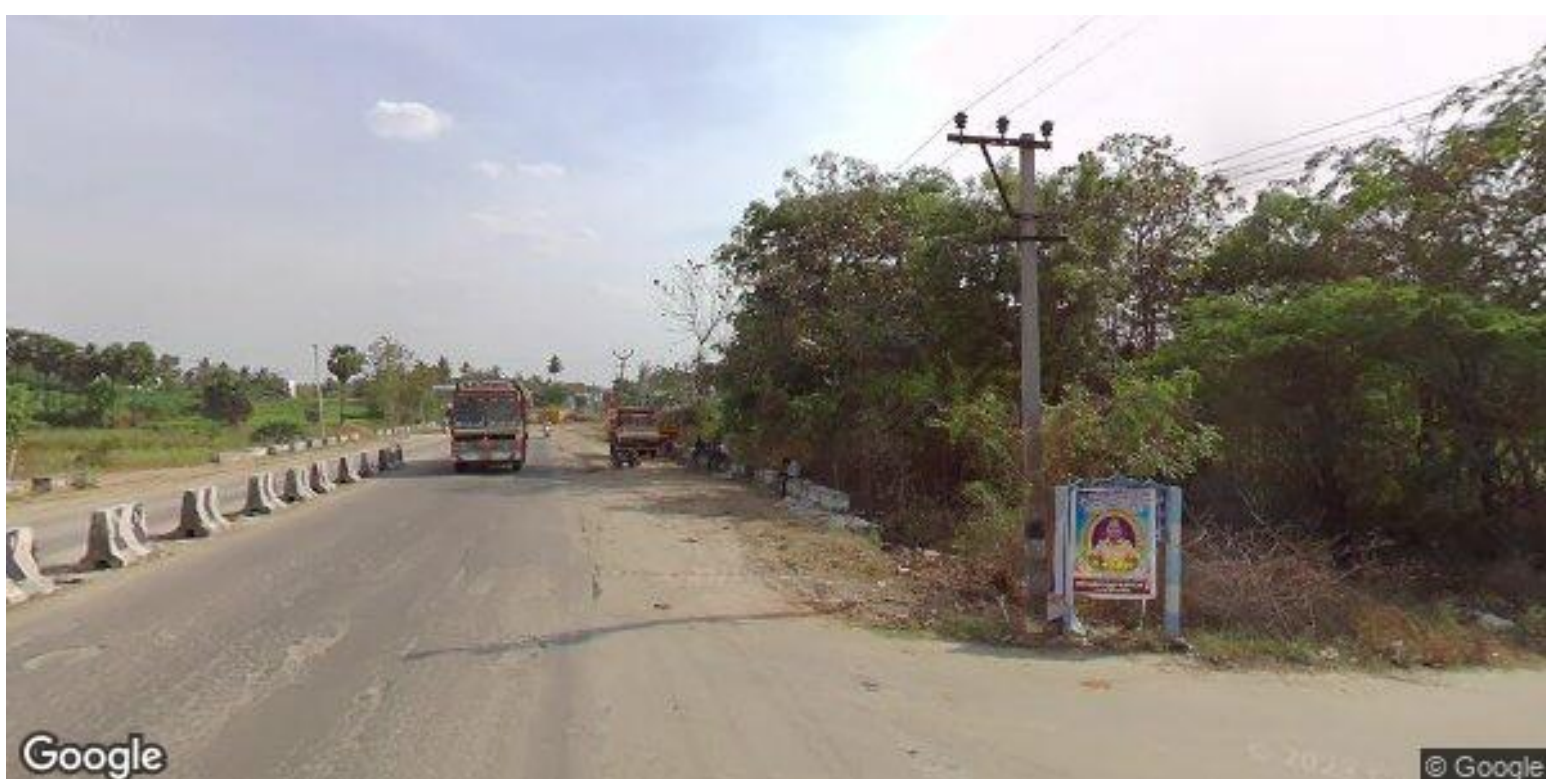


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58279, 77.90658



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63833, 77.94818



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64171, 77.95184



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64243, 77.95198



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67858, 77.96834



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68915, 77.97734



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71921, 78.01133



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03878



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.93670, 78.05606



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93877, 78.05736



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93965, 78.05742



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.94007, 78.05886



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.93896, 78.06064
